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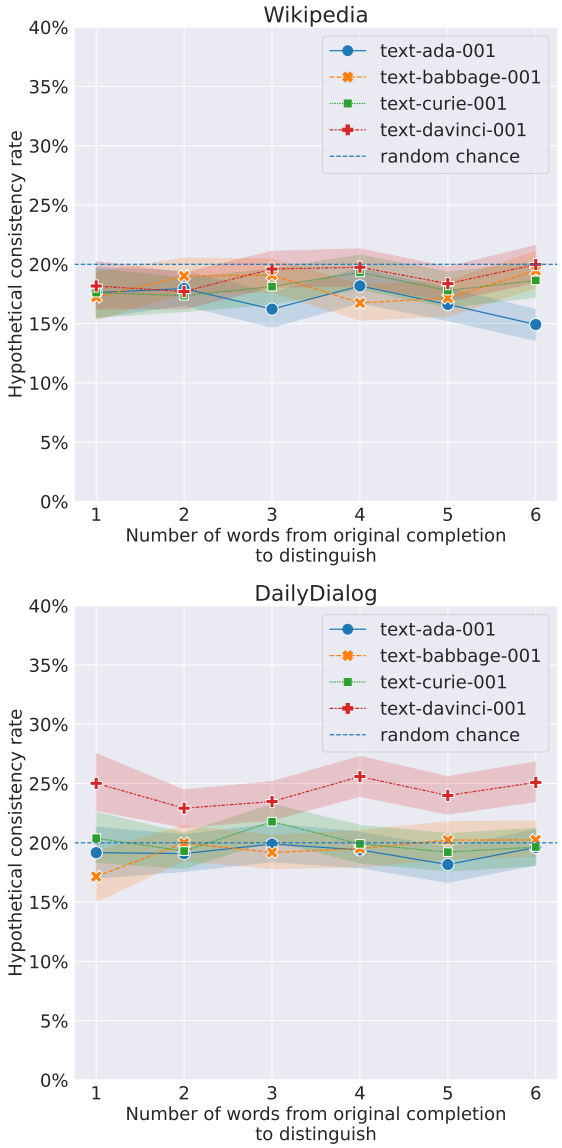
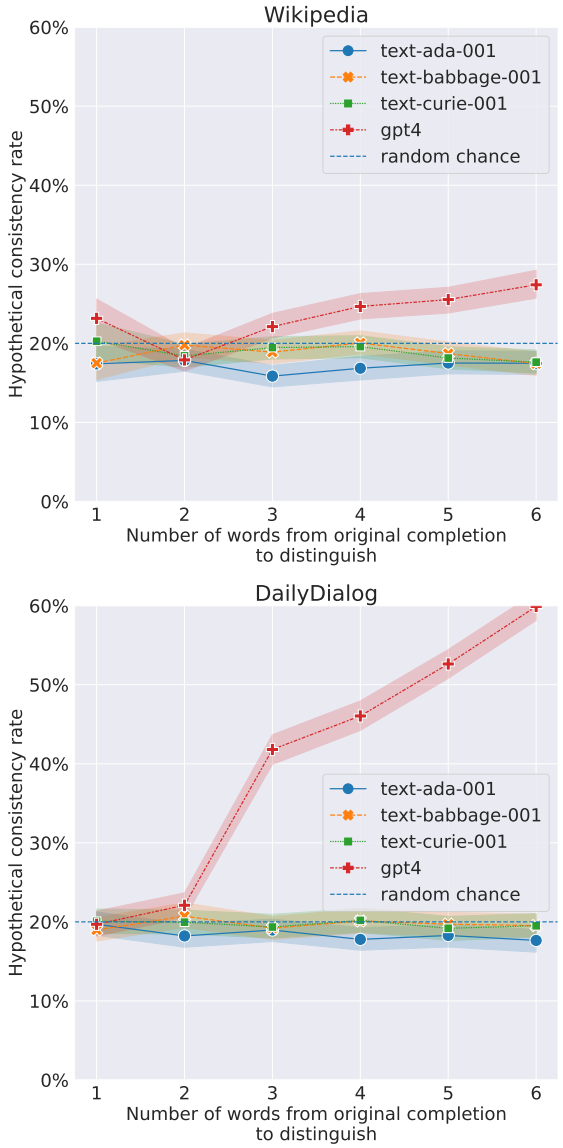
A Appendix



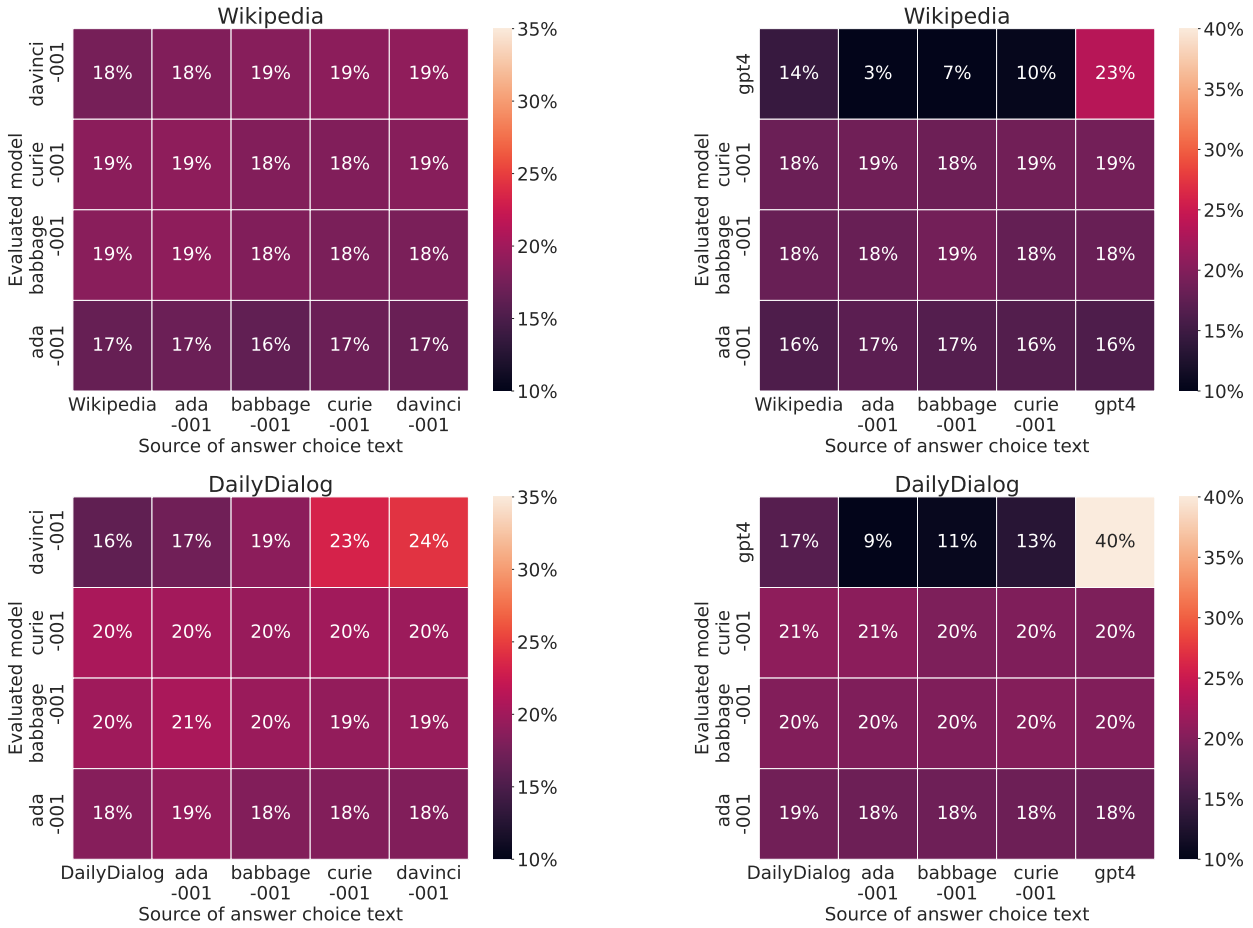
(a) Hypothetical consistency rates with answer choices generated by `ada-001`, `babbage-001`, `curie-001`, and `davinci-001`.



(b) Hypothetical consistency rates with answer choices generated by `ada-001`, `babbage-001`, `curie-001`, and `gpt4`.

Figure 6: Hypothetical consistency rates on multiple-choice hypothetical consistency prompts for the Wikipedia and DailyDialog datasets. Each multiple-choice prompt contains answer choices generated by the designated four models and an additional answer choice containing the actual continuation of the prompt in the dataset. Each line is the average taken across all k-shot prompts, for $k \in [1, \cdots, 10]$. The shaded region represents the 95% confidence interval computed with nonparametric bootstrapping. The label "number of words from original completion to distinguish" corresponds to the quantity m in Table 1.

Comparison of Hypothetical Consistency Against `davinci-001` and `gpt4` We also run the same hypothetical consistency experiments on `davinci-001` and `gpt4`. Hypothetical consistency rates for `davinci-001` versus the smaller models are shown in Figure 6a, where trends are similar, but `davinci-001` performs at random chance on Wikipedia, like all the other `-001` series models. On DailyDialog, however, `davinci-001`



(a) Comparing selected answers on multiple-choice prompts with answer choices generated by `ada-001`, `babbage-001`, `curie-001`, and `davinci-001`.

(b) Comparing selected answers on multiple-choice prompts with answer choices generated by `ada-001`, `babbage-001`, `curie-001`, and `gpt4`.

Figure 7: The proportion of the time that each model (`ada-001`, `babbage-001`, `curie-001`, and `davinci-001`) selects each possible answer choice when prompted with a hypothetical consistency prompt. Model outputs that could not be parsed into an answer choice are not included. The columns labeled "Wikipedia" and "DailyDialog" correspond to the answer choice containing the completion from the original dataset. Model outputs that could not be parsed into an answer choice are not included.

performs noticeably better than all the other model sizes. Similar trends occur in Figure 7a, where most models are equally likely to select each answer choice on the Wikipedia dataset, and `davinci-001` is more likely to select either its own or `curie-001`'s completion on the DailyDialog dataset.

In contrast, when `gpt4` is tasked with distinguishing its own completions from those of `ada-001`, `babbage-001`, `curie-001`, and the dataset, `gpt4` performs notably better than both `davinci-001` and `davinci-003` on DailyDialog, reaching 59.9% hypothetical consistency when the number of words to distinguish is 6 (Figure 6b). However, its hypothetical consistency rate on Wikipedia is comparable to that of `davinci-003` (Figure 2), ranging from 17.9% to 27.4%. Figure 7b also demonstrates that `gpt4` is significantly more likely to select its own completion than the other models are.

It is unclear why `gpt4` is more consistent on DailyDialog than previous models of similar capacity (*i.e.* `davinci-001` and `davinci-003`). Little is known about `gpt4`'s architecture or training, aside from its multimodal abilities and training via reinforcement learning from human feedback (RLHF, OpenAI, 2023).